

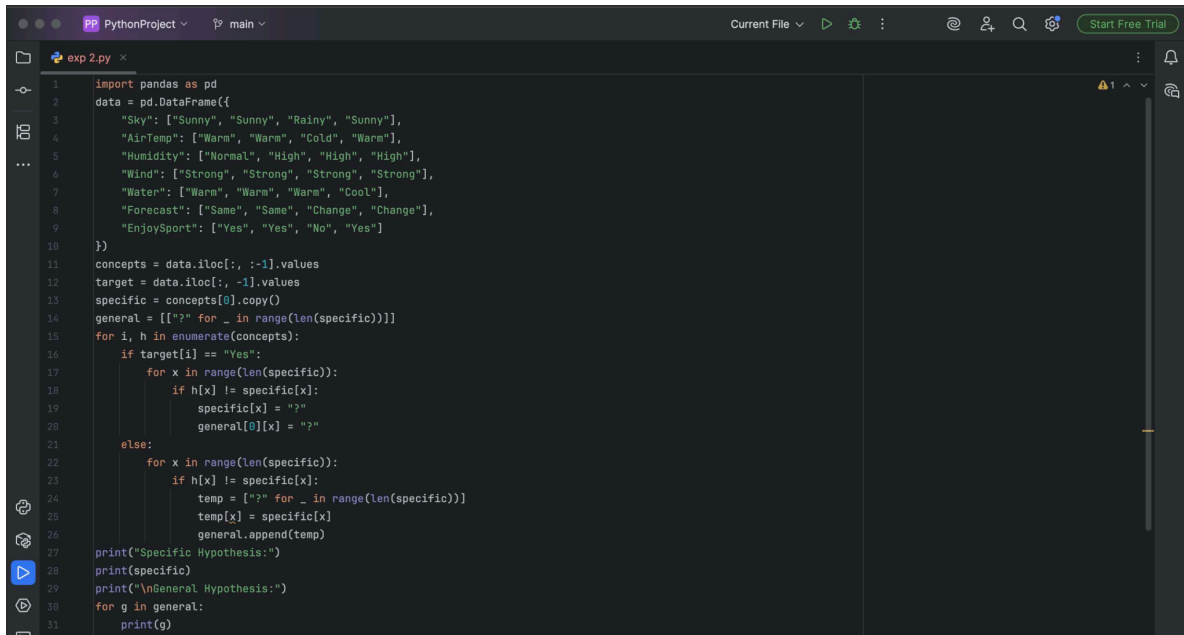
EXPERIMENT - 2

Aim: To find the set of all hypotheses consistent with the given training examples using the Candidate-Elimination algorithm.

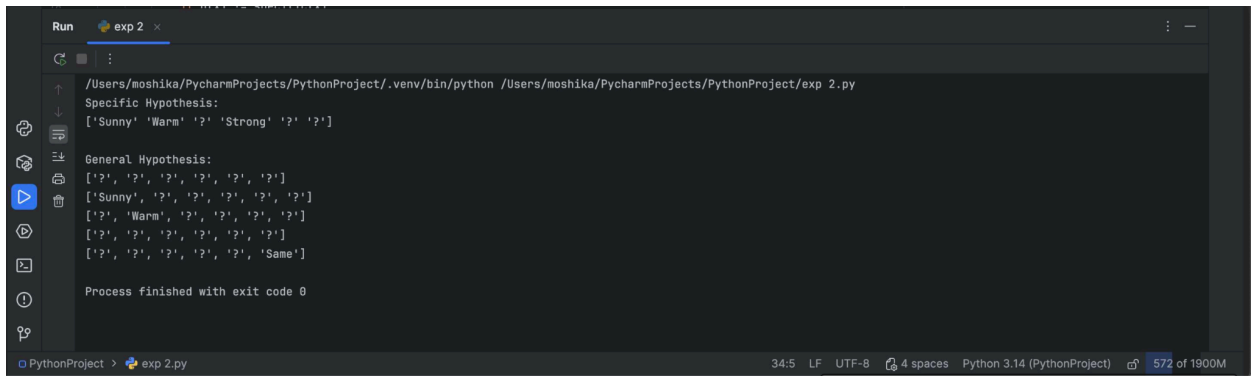
Procedure:

1. Load the training dataset from the CSV file.
2. Initialize S with the most specific hypothesis.
3. Initialize G with the most general hypothesis.
4. Read each training example.
5. For a positive example, generalize S if required.
6. Remove inconsistent hypotheses from G.
7. For a negative example, specialize G if required.
8. Remove inconsistent hypotheses from S.
9. Display the final S and G boundaries.

Output:



```
1 import pandas as pd
2 data = pd.DataFrame({
3     "Sky": ["Sunny", "Sunny", "Rainy", "Sunny"],
4     "AirTemp": ["Warm", "Warm", "Cold", "Warm"],
5     "Humidity": ["Normal", "High", "High", "High"],
6     "Wind": ["Strong", "Strong", "Strong", "Strong"],
7     "Water": ["Warm", "Warm", "Warm", "Cool"],
8     "Forecast": ["Same", "Same", "Change", "Change"],
9     "EnjoySport": ["Yes", "Yes", "No", "Yes"]
10 })
11 concepts = data.iloc[:, :-1].values
12 target = data.iloc[:, -1].values
13 specific = concepts[0].copy()
14 general = [["?" for _ in range(len(specific))]]
15 for i, h in enumerate(concepts):
16     if target[i] == "Yes":
17         for x in range(len(specific)):
18             if h[x] != specific[x]:
19                 specific[x] = "?"
20                 general[0][x] = "?"
21     else:
22         for x in range(len(specific)):
23             if h[x] != specific[x]:
24                 temp = ["?" for _ in range(len(specific))]
25                 temp[x] = specific[x]
26                 general.append(temp)
27 print("Specific Hypothesis:")
28 print(specific)
29 print("\nGeneral Hypothesis:")
30 for g in general:
31     print(g)
```



```
Run exp 2 x
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 2.py
Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:
['?', '?', '?', '?', '?', '?']
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']

Process finished with exit code 0
```

PythonProject > exp 2.py 34:5 LF UTF-8 4 spaces Python 3.14 (PythonProject) 572 of 1900M

Result: The algorithm successfully produces the version space, representing all hypotheses consistent with the training examples.